

Homework Assignment 13

1. Problem 12-11, first part. Find the values of a and n for the R-5 propellant, and then find (A_b/A_t) . {Ans.: at the back of the text}
2. Problem 12-12, first part {Ans.: increases to 6.9 MPa}
3. Repeat the calculations of the axial pressure variation example at the end of the burn; that is, calculate $\dot{m}_2$, (p_2/p_1) , u_2 , M_2 , T_{02} and p_{02} . {Partial Ans.: $(p_2/p_1) \approx 0.98$, $M_2 = 0.14$ }
4. The regression rate of a solid propellant is $r = \frac{A}{T_1 - T_p} p_0^n$, where T_1 is 415 K, and $n = 0.716$.
 - (a) Calculate the values of Π_r and Π_p at propellant grain temperatures of 20°C and 45°C.
 - (b) With the initial propellant grain temperature at 20°C, the chamber pressure is 3 MPa. Integrating the defining equation of Π_p , determine the chamber pressure if the initial grain temperature is at 45°C. {Ans.: 6.73 MPa}